Shapley-Based Responsibility for Noisy–OR Network Diffusion Models

David Landsberg[1*]

[1*]Frazer-Nash Consultancy Ltd.

Corresponding author(s). E-mail(s): david.landsberg.cs@gmail.com;

Abstract

Quantifying how responsible a source is for spreading information is challenging when signals are noisy, users receive exposures from multiple sources, or contributions interact non-additively. We address this by modelling diffusion as a family of Noisy–OR exposure models, in which each post carries an independent propensity to influence an exposed user about a given topic. In a Noisy–OR model, multiple independent exposures each have their own chance of influencing a user. We define expected responsibility via the Shapley value and derive an exact closed-form expression specialised to Noisy–OR models. The method runs in quadratic time for every user in the network, representing a substantial improvement in efficiency compared to the factorial complexity of the brute-force Shapley definition, and avoids the approximation error of sampling methods by giving exact results. Experiments on the FibVID COVID–19 dataset demonstrate interpretable source-level rankings, temporal responsibility analysis, and scalability to realistically sized diffusion models.

Keywords: Information diffusion, attribution, diffusion cascades, causal attribution, Shapley value, cooperative game theory, social networks, Noisy–OR models

1 Introduction

How can we measure responsibility for spreading information about a given topic? By *responsibility* we mean the degree to which a spread can be said to depend on a given source (rather than intent or moral blame). Here, we focus on textual content disseminated online, where individuals are commonly exposed to multiple related claims originating from different sources. Formally defining and attributing responsibility in such settings is challenging for three reasons: (i) Content signals are noisy rather than binary. For example, a post might be assigned a probability in the range $[0, 1]$ of belonging to a given class (e.g. COVID-19 misinformation). (ii) Users often receive overlapping exposures from several sources, where they like or repost multiple similar posts on the same theme. (iii) The degree of responsibility of one source can depend on the presence of others, making contributions non-additive. For instance, if a user is exposed to similar content from independent sources, the user's dependence on that source is less than if that source was wholly responsible. A principled responsibility method is therefore required both for scientific analysis of diffusion dynamics, and for practical tools that summarise which sources are most responsible for a given spread.

A major challenge concerns an appropriate formal and principled definition of responsibility in this setting. A core requirement is that the resulting attributions be interpretable, admitting direct answers to questions of the form "what percentage of the spread is a given source responsible for?". Crucially, because content signals are subject to substantial measurement noise, such questions can only be answered *in expectation*. By *noise* we mean uncertainty in estimating whether a post actually contains the given information. For instance, automated misinformation detectors and fact-checking pipelines typically reflect this by returning graded scores rather than binary labels, capturing ambiguity of language, contextual framing, and the limitations of current semantic models. Similarly, fact-checking taxonomies such as PolitiFact [2] and Snopes [3] treat veracity of a post within a spectrum (e.g., *barely true*, *half true*, *mostly false*), rather than a dichotomy. A formal probabilistic model is therefore required to accommodate such noise and to attribute a notion of *expected responsibility* in a principled manner.

To address these challenges, we make the following contributions:

1. We introduce a probabilistic exposure framework for information diffusion, termed the *Noisy–OR Exposure model*, in which each post carries an independent propensity to inform, and overlapping exposures combine non-linearly.
2. To facilitate a principled and interpretable answer to questions of the form "What percentage of the spread is [this account] responsible for?", we formalise *expected responsibility* by constructing a cooperative game induced by the Noisy–OR exposure model and applying the Shapley value, yielding principled attributions that satisfy the axioms of efficiency, symmetry, null player, and additivity.
3. We derive an exact closed-form Shapley expression for Noisy–OR games. This yields an $O(n^2)$-time, $O(n)$-space algorithm for computing all post-level attributions for a user exposed to n posts, avoiding factorial enumeration [1] while returning exact rather than Monte Carlo estimates.
4. To demonstrate the practical utility of the method, we conduct an empirical study on the FibVID COVID–19 misinformation dataset, demonstrating that the method yields interpretable source rankings, supports temporal responsibility analysis, and is computationally practical for realistically sized problems.

To the best of our knowledge, existing work does not provide a Shapley-based attribution framework specialised to probabilistic Noisy–OR diffusion models together with an exact closed-form quadratic-time algorithm.

The remainder of the paper proceeds as follows. We review previous and related work in Section 2. Section 3 introduces the exposure-model formalism and the Noisy–OR mechanism used to represent noisy, overlapping exposures, and shows how user-level exposure functions induce cooperative games. Section 4 defines our responsibility measure via the Shapley value, and introduces three baseline contribution measures for comparison. Section 5 derives the specialised Shapley equation for Noisy–OR games and presents the efficient algorithm. Section 6 reports an empirical study on the FibVID COVID–19 misinformation dataset, demonstrating how we can use our measure to describe source-level rankings, network structure among significant actors, and temporal responsibility dynamics. Discussion and conclusions are presented in Sections 7 and 8, respectively.

2 Related Work

Related work ranges from empirical studies on information exposure and receptivity, development of algorithms for influence maximisation and diffusion, and the general study of Shapley and causal attribution.

Influence maximisation and diffusion

The development of methods to maximise influence and diffuse information in social networks have been extensively studied. Threshold and cascade-based models, from Granovetter's threshold framework [18] to scalable influence-maximisation algorithms [19], linear threshold models [13] focus on selecting seed sets or interventions to optimise expected reach. Budak et al. [14] develop strategies for selecting optimal initiators and intervention policies, including group disbanding, to limit misinformation spread within echo chambers. Narayanam et al. [15] propose Shapley value-based methods for " discovering a small subset of influential players in a given social network, to perform a certain task of information diffusion", but do not address the attribution problems discussed in this paper.

Attribution

The Shapley value is the canonical solution concept in cooperative game theory for fairly distributing a coalition's total value according to axioms of efficiency, symmetry, null player, and additivity [7]. Surveys of cooperative game theory emphasise that the Shapley value's axiomatic character makes it a natural candidate for responsibility-like allocations [22]. In explainable machine learning, Shapley-based methods such as `shap` assign feature-level importance scores to individual predictions, relying on a cooperative game over input variables [21], and are a prominent concept in the field of explanatory AI. Shapley values have been applied to related attribution problems in social networks, including assigning influence scores to authors, citation webs, and other collaborative agents where contributions interact in non-additive ways [16]. However, these applications do not instantiate a concrete noisy exposure model on large social graphs, and existing complexity results do not directly exploit the algebraic structure of Noisy–OR value functions to obtain exact quadratic-time algorithms.

Finally, there are (non-Shapley based) approaches to attribution influenced by concepts of causal explanation [24], and responsibility, and blame [25]. Here, computing responsibility is limited to the deterministic case, splits responsibility equally amongst independently sufficient causes, and more generally requires knowledge of "the minimal number of changes that have to be made" for a counterfactual dependence to hold between the effect and the cause, which in our noisy context is unknown.

Information exposure and receptivity

Several studies measure users' exposure and receptivity to misinformation by investigating user behaviour. Vosoughi et al. [20] show that false news tends to spread farther, faster, and more broadly than true content on Twitter. Tokita et al. [8] combine Twitter exposure data with real-time survey responses for 139 viral posts, demonstrating that ideologically extreme users are exposed to and more likely to accept false news earlier than others. Mosleh et al. [9] quantify exposure to misinformation by political elites, finding that exposure predicts poorer sharing quality and exhibits a political skew. Avram et al. [10] design an experiment to show that high social engagement cues (likes/shares) increase susceptibility to low-credibility content. Del Vicario et al. [11] analyse Facebook cascades and show that misinformation spreads within polarized echo chambers, forming deeper and wider diffusion structures than scientific content. Pierri et al. [12] compare Twitter diffusion network topologies, finding that misleading-news cascades have larger weakly connected components and higher clustering than mainstream ones. Lazer et al. [17] survey the empirical and methodological challenges in studying fake news, emphasising the need for rigorous, transparent measures of exposure and influence in complex online environments; serving to motivate our current measure.

Taken together, these strands leave open the problem addressed in this paper: how to allocate expected responsibility in diffusion settings where users receive noisy, overlapping exposures from multiple posts and sources. The key distinction from generic Shapley-based attribution methods is that our cooperative game has Noisy–OR structure. Exploiting this structure yields an exact closed-form Shapley expression for post-level responsibility, allowing all responsibilities for a recipient to be computed in polynomial time rather than by factorial enumeration or Monte Carlo approximation. Thus, the contribution is not simply the use of the Shapley value, but the identification of a tractable responsibility-allocation problem within noisy network diffusion.

3 Theoretical Background

In this section we present our definition of Noisy–OR exposure models, and connect them to cooperative game theory. This provides a foundation for our definition of expected responsibility in the next section.

3.1 Exposure Models

We begin with a general definition of an *exposure model* (aka information diffusion model), which specifies who was exposed to which piece of content, when, and which account was the root-cause of that exposure. In this work, exposure models are used solely to characterise relationships between *sources*—the original authors of posts that initiate a diffusion (aka root causes) — and *recipients*, defined as downstream users who interact with that content either directly or indirectly. To support root–cause analysis, we deliberately abstract away from the internal structure of reposting cascades: intermediary users are collapsed, so that if user a originates a post, user b reposts it, and user c reposts b's repost, the only information retained is that a's content exposed both b and c to the content. A key feature of this abstraction is that it preserves root-causal provenance. Finally, we assume that each model is associated with a fixed latent type (e.g. COVID–19 misinformation), allowing exposures to be interpreted consistently as instances of exposure to information of that type.

Definition 1 (Exposure Model) An *Exposure Model* is a directed multigraph $M = (V, T, E, \mathcal{A}, \rho)$ where:

- V is a finite set of *vertices* representing network *users*;
- T is a finite set of *timestamped IDs*;
- $E \subseteq V \times V \times T$ is the set of directed edges, called *exposure events*;
- $\mathcal{A}$ is the set of *source-post IDs*;
- $\rho : E \to \mathcal{A}$ assigns to each exposure event the source-post ID associated with it.

Readings. Informally, an exposure model represents the spread of information on a topic in relation to the sources who began propagating it, and the downstream recipients. An edge $(s, r, t) \in E$ represents an *exposure event* in which *recipient* r engages with content at time t which was originally posted by *source* s (e.g. by reposting a post of the given source, or by reposting someone else's repost of the given source, etc.). Intuitively, the direction of the arrow reflects the direction of information flow. The function ρ maps an exposure event to the corresponding source-post ID in $\mathcal{A}$, which uniquely identifies the source's post. Parallel edges between the same pair of individuals are permitted, and correspond to different exposure events. Hereonin, we will refer to source-post IDs interchangeably with their corresponding posts, and talk of users being exposed to the posts in $\mathcal{A}$ for simplicity.

Assumptions. For clarity, we adopt the following formal assumptions:

1. *Irreflexivity.* Exposures are never self-directed, i.e. $\forall (s, r, t) \in E,\ s \neq r$, so a user is not treated as exposing themselves to their own content.
2. *Timestamp uniqueness and total ordering.* If $(s, r, t) \in E$ and $(s', r, t') \in E$ with $s \neq s'$, then $t \neq t'$. Equivalently, for each recipient r, exposure events admit a total ordering by timestamp. We interpret an *exposure* as a discrete engagement event—such as a repost, like, or other interaction—through which a recipient engages with content; on platforms of interest these events occur sequentially and are typically recorded with sufficiently fine-grained timestamps. When equal timestamps occur in practice, ties may be broken using some heuristic, or arbitrarily.
3. *Unique origin per source-post ID.* Source-post IDs are associated with a unique source, i.e. $\forall s, s' \in V,\ \forall r, r' \in V,\ \forall t, t' \in T,\ \big(\rho(s, r, t) = \rho(s', r', t')\big) \Rightarrow s = s'$. Operationally, this matches platforms where each post has a globally unique ID tied

source	recipient	post ID	time
s_1	u_1	p_1	t_1
s_2	u_1	p_2	t_2

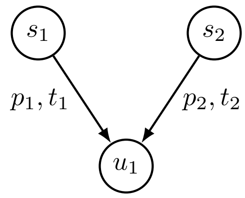


Fig. 1 Toy exposure model with two independent sources. Left: tabular representation, Right: equivalent directed graph representation.

to its original author, while downstream interactions create new exposure events referencing that ID for provenance.

Utilities. It will be convenient to have the following functions available. For an exposure model $M = (V, T, E, \mathcal{A}, \rho)$:

- $\mathrm{ex} : V \times 2^{\mathcal{A}} \to 2^{\mathcal{A}}$, $\mathrm{ex}(r, A) = \{\, a \in A : \exists s, t.\, (s, r, t) \in E \wedge \rho(s, r, t) = a \,\}$ — posts in A to which recipient r was exposed;
- $\mathrm{art} : V \to 2^{\mathcal{A}}$, $\mathrm{art}(s) = \{\, a \in \mathcal{A} : \exists r, t.\, (s, r, t) \in E \wedge \rho(s, r, t) = a \,\}$ — posts posted by source s;
- $\mathrm{src} : \mathcal{A} \to V$, $\mathrm{src}(a) = s$ iff $\exists r, t.\, (s, r, t) \in E \wedge \rho(s, r, t) = a$ — unique source of post a;
- $\mathcal{S} = \{\mathrm{src}(a) \mid a \in \mathcal{A}\} \subseteq V$ — set of all sources.

Example 1 We present a toy example of an exposure model. Here, we model a simple case where a recipient/user (called u_1) reposts from each of two independent sources (called s_1 and s_2) at different times (t_1 and t_2, letting the IDs of the source-posts be p_1 and p_2). Accordingly, $V = \{s_1, s_2, u_1\}$ are the three users, $T = \{t_1, t_2\}$ are the timestamps, $E = \{(s_1, u_1, t_1), (s_2, u_1, t_2)\}$ are the exposure events, and $\mathcal{A} = \{p_1, p_2\}$ are the source–post IDs. where $\rho(s_1, u_1, t_1) = p_1$ and $\rho(s_2, u_1, t_2) = p_2$. A corresponding tabular and graphical representation is provided in Figure 1. For this model: $\mathrm{ex}(u_1, \mathcal{A}) = \{p_1, p_2\}$, $\mathrm{art}(s_1) = \{p_1\}$, $\mathrm{art}(s_2) = \{p_2\}$, $\mathrm{src}(a_1) = s_1$, $\mathrm{src}(a_2) = s_2$, and $\mathcal{S} = \{s_1, s_2\}$.

3.2 Noisy–OR Exposure Models

We now extend exposure models to accommodate settings in which information signals are noisy rather than deterministic. Each post $i \in \mathcal{A}$ is assigned a *causal propensity* $w_i \in (0, 1]$, interpreted as the probability that exposure to post i causes a recipient to be informed about some latent topic. We write $a_i := 1 - w_i$ for the complement, representing the probability that exposure to post i does *not* cause a recipient to be informed. If a user is exposed to multiple posts $A \subseteq \mathcal{A}$, we assume that each post in A acts as an independent causal mechanism with respect to whether the user is informed. Under this assumption, the probability that at least one post in A causes the recipient to be informed is given by the standard Noisy–OR formula $1 - \prod_{i \in A}(1 - w_i)$, as used for independent causal mechanisms in probabilistic graphical models [23], and represents a parsimonious baseline in the absence of interaction data.

source	recipient	post-ID	time	weight
s_1	u_1	p_1	t_1	0.5
s_2	u_1	p_2	t_2	0.5

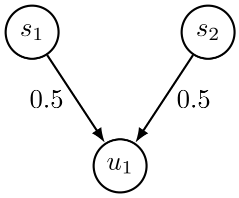


Fig. 2 Toy Noisy–OR exposure model with two independent sources. Left: tabular representation with weights. Right: equivalent directed multigraph view with edge-weights

Definition 2 (Noisy–OR Exposure Model) Let $M = (V, T, E, \mathcal{A}, \rho)$ be an exposure model and let $w : \mathcal{A} \to (0, 1]$ be a given weight function. A *Noisy–OR exposure model* is the pair (M, w) together with the following construction.

For each user $u \in V$, define the *user exposure function* $v_u : 2^{\mathcal{A}} \to [0, 1]$ by

$$v_u(A) = 1 - \prod_{i \in \mathrm{ex}(u,A)} (1 - w_i) \qquad (A \subseteq \mathcal{A}). \tag{1}$$

Define the *global exposure function* $\mathbf{v} : 2^{\mathcal{A}} \to \mathbb{R}_{\geq 0}$ by

$$\mathbf{v}(A) = \sum_{u \in V} v_u(A) \qquad (A \subseteq \mathcal{A}). \tag{2}$$

We adopt the empty-product convention $\prod_{i \in \varnothing}(1 - w_i) = 1$; hence $v_u(\varnothing) = 0$ for all $u \in V$ and $\mathbf{v}(\varnothing) = 0$.

Readings. For a set of posts $A \subseteq \mathcal{A}$:

- $v_u(A)$ is the probability that user u is informed by *at least one* post in A;
- $\mathbf{v}(A)$ is the expected number of users who would be informed if only the posts in A were present.

When $A \subsetneq \mathcal{A}$, $\mathbf{v}(A)$ is a counterfactual estimate—the expected spread that would occur if all posts outside A had been removed. When $A = \mathcal{A}$, it yields the observed spread. We therefore interpret $\mathbf{v}(A)$ as the expected magnitude of the spread. We also call it the expected spread generated by A.

Example 2 We illustrate a simple Noisy–OR model (M, w). Let M be the exposure model of the previous example, and define the weight function by $w(p_1) = w(p_2) = 0.5$. Thus each post independently informs an exposed user with probability 0.5. The corresponding tabular and graphical representations are shown in Figure 2. Here, we assume there is some latent topic which the posts are about. For this model $\mathbf{v}(\mathcal{A}) = v_{u_1}(\mathcal{A}) = 1 - (1 - 0.5)(1 - 0.5) = 0.75$ so the expected number of users who have been informed about the topic is 0.75; equivalently, user u_1 is informed about the topic with probability 0.75.

3.3 Cooperative Games

We now make explicit the connection between noisy exposure models and cooperative game theory. It is first useful to establish some general terminology. A *cooperative* *game* is a pair (N, v) where $N = \{1, \ldots, n\}$ is a finite set of *players* and $v : 2^N \to \mathbb{R}_{\geq 0}$ is a *value function* satisfying $v(\varnothing) = 0$. A *coalition* is any subset $S \subseteq N$; the cases $S = \varnothing$ and $S = N$ are called the *empty coalition* and *grand coalition*, respectively. For a player $i \in N$ and a coalition $S \subseteq N \setminus \{i\}$, the *marginal contribution* of i to S is $v(S \cup \{i\}) - v(S)$.

Definition 3 (Noisy–OR Game) Each player $i \in N$ has an associated *weight* $w_i \in [0, 1]$, interpreted as its success probability, and complement $a_i := 1 - w_i$. The *Noisy–OR value function* is defined by

$$v(S) = 1 - \prod_{j \in S}(1 - w_j) = 1 - \prod_{j \in S} a_j, \qquad S \subseteq N. \tag{3}$$

By the empty–product convention $\prod_{j \in \varnothing} a_j = 1$, so $v(\varnothing) = 0$. A game (N, v) with value function of the form (3) is called a *Noisy–OR game*.

We now connect this definition to Noisy–OR exposure models. Fix a Noisy–OR exposure model (M, w), where $M = (V, T, E, \mathcal{A}, \rho)$ is an exposure model and w is the weight function. For each user $u \in V$, the associated exposure function v_u induces the Noisy–OR game $(\mathcal{A}, v_u)$ in which:

- the *players* are the posts in $\mathcal{A}$,
- a *coalition* $S \subseteq \mathcal{A}$ represents the set of posts present, and
- For each set of posts $S \subseteq \mathcal{A}$, $v_u(S)$ is the probability that user u is informed via S.

We identify the player set N with the post set $\mathcal{A}$, i.e. $N := \mathcal{A}$, and henceforth use the notation (N, v_u) when referring to this game in standard cooperative-game form. We also call the Noisy–OR game (N, v_u) a *user-level game*. We call $(N, \mathbf{v})$ the game for the Noisy–OR exposure model (M, w) and associated global exposure function $\mathbf{v}$. Finally, a *payment rule* assigns to each cooperative game (N, v) a vector $(\phi_i(N, v))_{i \in N} \in \mathbb{R}^N$, where $\phi_i(N, v)$ denotes the *payoff* allocated to player i.

4 Expected Responsibility in Noisy–OR Diffusion Models

In this section we develop our measure of *expected responsibility*. We first describe the properties we wish this rule to have and identify these with the Shapley axioms, and then present our definitions of expected responsibility. Finally, we illustrate the measure on some simple cases.

4.1 Shapley Value

Let N be the set of posts and v a given user's exposure function of a Noisy–OR game. We seek a payment rule ϕ satisfying the following properties.

- **Efficiency.** $\sum_{i \in N} \phi_i(N, v) = v(N)$.

- **Symmetry (Fairness).** If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all $S \subseteq N \setminus \{i, j\}$, then $\phi_i(N, v) = \phi_j(N, v)$.
- **Dummy (Null Player).** If $v(S \cup \{i\}) = v(S)$ for all $S \subseteq N$, then $\phi_i(N, v) = 0$.
- **Additivity.** For any family of games (N, v_u) indexed by $u \in V$,

$$\phi_i\left(N, \sum_{u \in V} v_u\right) = \sum_{u \in V} \phi_i(N, v_u),$$

where $(\sum_{u \in V} v_u)(S) = \sum_{u \in V} v_u(S)$.

These axioms capture natural desiderata for a payoff rule describing expected responsibility of a post in noisy diffusion settings: Efficiency ensures that expected responsibility is conserved; the sum of post-level expectations equals the expected total spread. Symmetry encodes fairness; posts with identical roles receive identical degrees of expected responsibility. The dummy property ensures that inert posts receive zero expected responsibility. Additivity enables aggregation across users, so that a post's overall expected responsibility equals the sum of its expected responsibilities in each user-level game. Shapley proved that there is a unique payment rule satisfying these four axioms [7], known as the *Shapley value*.

Definition 4 (Shapley value) For a cooperative game (N, v) with $|N| = n$, the *Shapley value* is the payment rule

$$\phi(N, v) = (\phi_1(N, v), \ldots, \phi_n(N, v)),$$

where the *Shapley value of player* $i \in N$ is

$$\phi_i(N, v) := \frac{1}{n!} \sum_{\pi \in \Omega} \big[v(\mathrm{Pre}_i(\pi) \cup \{i\}) - v(\mathrm{Pre}_i(\pi))\big]. \tag{4}$$

Here Ω is the set of all $n!$ permutations of N, and $\mathrm{Pre}_i(\pi)$ denotes the set of players preceding i in permutation π. Thus $\phi_i(N, v)$ is the expected marginal contribution of i when players join the coalition in a uniformly random order.

An equivalent form [7] is

$$\phi_i(N, v) = \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{\binom{n-1}{k}} \sum_{S \in \mathcal{P}_k(N \setminus \{i\})} \big(v(S \cup \{i\}) - v(S)\big), \tag{5}$$

where $\mathcal{P}_k(S) := \{\, T \subseteq S : |T| = k \,\}$ denotes the family of k-element subsets of S, with the convention $\mathcal{P}_0(S) = \{\varnothing\}$. We call Equation 4 the *permutation form* of the Shapley value, and Equation 5 the *subset form*.

4.2 Post-level Expected Responsibility

We now formally connect Shapley values to post-level responsibility.

Definition 5 (post-level expected responsibility) Let (M, w) be a Noisy–OR exposure model, $N := \mathcal{A}$ be the set of posts associated with M, $\mathbf{v}$ its associated a global exposure function, and (N, v) its associated game. The *expected responsibility* of a post is defined as its Shapley value over the associated game:

$$C_{\text{art}-\text{resp}}(i) = \phi_i(N, \mathbf{v}) \quad (i \in N) \tag{6}$$

Note. $\phi_i(N, \mathbf{v}) = \phi_i(N, \sum_{u \in V} v_u) = \sum_{u \in V} \phi_i(N, v_u)$ (following the definition of the global exposure function, and the additivity property of the Shapley value). This means that post-level responsibility reduces to computing the Shapley value for a series of user-level games. We now continue with the definition of source-level responsibility.

We establish an intuition for how $C_{\text{art-resp}}$ connects to the Shapley formula stated in Equation 4. Here, the expected responsibility of a post is interpreted as its expected marginal contribution to the spread, taken over all logically possible orders in which posts could be posted, and under the assumption that each order is equally likely. This interpretation is particularly natural when the precise causal relationships between source posts is unknown.

4.3 Source-level Expected Responsibility

Definition 6 (Source-level expected responsibility) For a source $s \in \mathcal{S}$ with associated posts $\text{art}(s) \subseteq N$, the *expected responsibility of a source* is defined as the aggregation of post-level Shapley values under the partition of posts induced by sources:

$$C_{\text{resp}}(s) = \sum_{i \in \text{art}(s)} \phi_i(N, \mathbf{v}). \tag{7}$$

Its *percentage expected responsibility* is

$$C_{\text{resp}\%}(s) = \frac{C_{\text{resp}}(s)}{\mathbf{v}(N)} 100. \tag{8}$$

Because a source acts only through the posts it posts, the expected responsibility of a source is naturally identified with the sum of the expected responsibilities of the posts they posted. The definitions of source and post-level responsibility reflect this, given $C_{\text{resp}}(s) = \sum_{i \in \text{art}(s)} C_{\text{art}-\text{resp}}(i)$. The intuition here is that the responsibility of a source is directly tied to their actions, where an action is posting a post. This aggregation is principled: by additivity of the Shapley value, summing post-level responsibilities preserves the total expected spread and avoids double counting when multiple posts contribute to the same exposures. The percentage responsibility $C_{\text{resp}\%}(s)$ therefore admits a interpretation as the fraction of the expected spread that is attributable to source s, and the efficiency property of the Shapley value ensures that $\sum_{s \in \mathcal{S}} C_{\text{resp}\%}(s) = 100$. Accordingly, statements of the form "source s was responsible for x% of the spread" are meaningful in expectation. We refer to $C_{\text{resp}}(s)$ computed using the full Noisy–OR exposure model as the *global responsibility* of source s.

4.4 Dynamic Source-level Expected Responsibility

We distinguish the global responsibility of a source from its *dynamic responsibility*, described by a *dynamic responsibility matrix*. This is a matrix whose rows correspond to sources, whose columns correspond to timestamps, and whose entries record the change in expected responsibility for each source relative to the preceding timestamp. This representation makes it possible to track how expected responsibility accrues over time, to identify when individual sources begin to dominate a diffusion, and to distinguish early causal drivers from later amplifiers.

Definition 7 (Dynamic responsibility matrix) Let S be a finite set of sources and let $\{t_1 < t_2 < \cdots < t_T\}$ denote discrete timestamps. For a noisy exposure model and a responsibility function $C_{\mathrm{resp}}(s; t_\ell)$ giving the expected responsibility of source $s \in S$ up to timestamp t_ℓ, the *dynamic responsibility matrix* is the matrix $\mathbf{R} \in \mathbb{R}^{|S| \times T}$ with entries

$$\mathbf{R}_{s,1} = C_{\mathrm{resp}}(s, t_1), \tag{9}$$

$$\mathbf{R}_{s,\ell} = C_{\mathrm{resp}}(s, t_\ell) - C_{\mathrm{resp}}(s, t_{\ell-1}), \quad 2 \le \ell \le T. \tag{10}$$

The dynamic responsibility matrix provides an intuitive and interpretable representation of how responsibility accumulates and shifts between sources over time. We assume that each user's engagement events are totally ordered in time, as required by the exposure model. When timestamps are not strictly ordered in the available raw data, ties may be broken arbitrarily, at the expected cost of introducing a minor modelling approximation.

4.5 Comparison with Basic Measures

To clarify the behaviour of causal responsibility C_{resp}, we compare it with two simple baseline notions of *causal influence*, which can be interpreted as measuring the degree of causal necessity, and causal sufficiency respectively. Accordingly, we describe their intuitive interpretations, and contrast their outputs with expected responsibility on toy examples.

Definition 8 (Independent Causal Sufficiency) For a source $s \in \mathcal{S}$ with posts $\mathrm{art}(s) \subseteq \mathcal{A}$,

$$C_{\mathrm{suff}}(s) = \mathbf{v}(\mathrm{art}(s)).$$

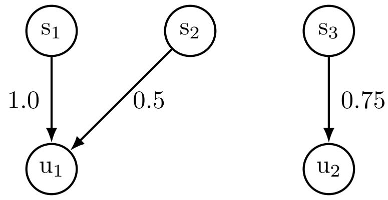
This measures the expected magnitude of the spread if only s's posts had been posted.

Definition 9 (Degree of Causal Necessity) For a source $s \in \mathcal{S}$,

$$C_{\mathrm{nec}}(s) = \mathbf{v}(\mathcal{A}) - \mathbf{v}(\mathcal{A} \setminus \mathrm{art}(s)).$$

This measures the expected reduction in spread if s had not posted any of their posts.

We present these measures as they have an intuitive connection to causality. Although C_{suff} and C_{nec} can serve as useful indicators of influence, they embody different and often conflicting intuitions, and may rank sources in substantially different ways to C_{resp} even on small examples. Further, if interpreted as capturing responsibility, they produce allocations that may reasonably be regarded as inadequate to the task, as the following example illustrates:



	Uniform				Noisy			
Measure	s1	s2	s3	Ratio	s1	s2	s3	Ratio
C_{suff}	1	1	1	1:1:1	1	$\frac{1}{2}$	$\frac{3}{4}$	4:2:3
C_{nec}	0	0	1	0:0:1	$\frac{1}{2}$	0	$\frac{3}{4}$	2:0:3
C_{resp}	$\frac{1}{2}$	$\frac{1}{2}$	1	1:1:2	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	3:1:3

Fig. 3 Toy Comparison of Basic Measures

Example 3 Consider the toy example in Figure 3. The example specifies an exposure model represented as a multigraph, with sources labelled s_1, s_2, s_3 and recipients labelled u_1, u_2. Two redundant sources (s_1, s_2) expose one user and s_3 acts independently on a second user u_2. For the model we consider two interpretations. In the *uniform interpretation* we set all weights to 1, corresponding to certainty that every exposure caused the recipient to be misinformed. In the *noisy interpretation* we retain the given edge weights, where each weight represents the estimated probability that exposure caused the recipient to be misinformed. For each interpretation and measure, we also provide the scores computed by that measure, along with the ratio between those scores. We discuss the example as follows.

Under the uniform interpretation, C_{resp} is the unique measure to identify s_3 as the most responsible, and to attribute some positive degree of responsibility to the remaining sources. We think this is the correct result; s_3 is most responsible because s_3 misinformed one recipient by themselves, but the other two sources were each redundant in misinforming the remaining recipient. In addition, although the remaining sources were each redundant, they were each causally sufficient to misinform one user so qualify for a positive attribution which should be reflected by the score.

Following this, we think the other measures are inadequate to the task of describing degrees of causal responsibility,: the measure of causal sufficiency scores all three sources equally (inflating the roles of s_1 and s_2), and the measure of causal necessity attributes zero responsibility to s_1 and s_2 (deflating their roles). Accordingly, C_{resp} gives a distribution of scores that better reflects responsibility: s_1 and s_2 receive positive but reduced responsibility, while s_3 is identified as the most responsible source.

We now turn to the noisy interpretation. This case demonstrates how the measures can provide different rank orderings: C_{suff} ranks s_1 as most responsible, C_{nec} ranks s_2 as most responsible, and C_{resp} ranks s_1 and s_3 as equally most responsible. The practical implication for larger datasets is that such discrepancies may compound, potentially obscuring which actors are most responsible.

Finally, we note that one may construct a wide range of heuristics that loosely approximate the Shapley value. For example, normalised split rules that involve dividing each exposure weight by the recipient's total incoming weight. However, following Shapley's proof no such alternative rule can satisfy all the four Shapley axioms presented in Section 4.1 unless it coincides with the Shapley value itself [7].

5 Algorithm and Complexity

In this section we present an efficient method for computing expected responsibility. By construction, this reduces to computing the Shapley value in Noisy–OR games.

5.1 Computing the Shapley Value

Recall that C_{resp} is obtained by allocating Shapley payoffs to individual posts and then aggregating by source. C_{resp} can consequently be thought of as a post-hoc aggregation of Shapley values. Hence the computational core is the evaluation of $\phi_i(N, v)$ for each post i. By the additivity axiom of the Shapley value,

$$\phi_i(N, \mathbf{v}) = \sum_{u \in V} \phi_i(N, v_u),$$

so it suffices to compute $\phi_i(N, v_u)$ for each user-level Noisy–OR game (N, v_u). We therefore first derive a closed-form expression for the Shapley value specialised to Noisy–OR games.

Proposition (Shapley Equation for Noisy–OR Games) *Let $N := [n]$ and let (N, v) be a Noisy–OR game with post weights w_i and complements $a_i = 1 - w_i$. Then the Shapley value of any player $i \in N$ satisfies*

$$\phi_i = \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{e_k^{-i}}{\binom{n-1}{k}}, \tag{11}$$

where e_k^{-i} denotes the k-th elementary symmetric polynomial of the vector $(a_j)_{j \in [n] \setminus \{i\}}$.

Proof The result follows by substituting the Noisy–OR marginal contribution into the binomial form of the Shapley value and recognising the resulting inner sums as elementary symmetric polynomials. A complete derivation is given in Proposition 2 in Appendix A. □

Equation (11) replaces the factorial sum over permutations with a structured expression involving symmetric polynomials. The remaining task is to evaluate these quantities efficiently.

Proposition *Let (N, v) be a Noisy–OR game, letting $n = |N|$. There exists an algorithm that computes the exact Shapley vector $\phi(N, v)$ as given in Equation (11) in $O(n^2)$ time and $O(n)$ space.*

Proof See Proposition 5 in Appendix A. In outline, Algorithm 1 constructs the required elementary symmetric polynomials iteratively, avoiding explicit enumeration of coalitions and requiring only $O(n^2)$ arithmetic updates with linear storage. □

Accordingly, computing global responsibility reduces to evaluating one Noisy–OR Shapley game per user. If user u is exposed to n_u distinct posts, the cost per user is $O(n_u^2)$ time and $O(n_u)$ space. Summing over users yields an overall runtime of $O\big(\sum_{u\in V} n_u^2\big)$, and hence the coarse worst–case bound $O(|V|\, n_{\max}^2)$ where $n_{\max} = \max_{u\in V} n_u$. Since each user-level Shapley game is independent, the computation is embarrassingly parallel [26]. We now discuss practical tractability. In realistic diffusion datasets we can expect n_u to be small. For example, in the FibVID dataset the maximum exposure sequence length is $n_u = 63$, making the bound easily tractable in practice.

The primary contribution of this section is the reduction to an efficient Shapley computation, while the algorithmic details are technical, the full pseudo-code and supporting proofs are deferred to Appendix A.

5.2 Computing the Dynamic Responsibility Matrix

We compute the *dynamic responsibility matrix* of definition 7 using the following method: For each recipient $u \in V$, let $a_1 \prec a_2 \prec \cdots \prec a_{n_u}$ denote the posts to which u is exposed, ordered by timestamp. For each prefix $A_k := \{a_1, \ldots, a_k\}$, we compute the Shapley value $\phi(\, A_k, v_u\,)$ using Algorithm 1. post-level values are then aggregated by source, yielding a vector of source responsibilities at each time step for each user. Stacking these vectors across all sources and time steps yields the dynamic responsibility matrix.

We now discuss the time complexity of this construction. For a fixed user u, computing dynamic responsibility requires evaluating the Shapley value for prefix sizes $k = 1, 2, \ldots, n_u$. Since Algorithm 1 runs in $O(k^2)$ time on a k-player Noisy–OR game, the total cost per user is

$$\sum_{k=1}^{n_u} O(k^2) = O(n_u^3).$$

Summing over all users yields an overall runtime of

$$O\left(\sum_{u\in V} n_u^3\right).$$

We now discuss the space complexity of this construction. At any time, Algorithm 1 stores only $O(k)$ intermediate values for a k-post prefix. The dynamic responsibility matrix itself is typically sparse: a source's responsibility changes only at time steps where that source appears in a user's exposure sequence. Thus memory usage is dominated by the number of nonzero (source, time) entries as opposed to a dense matrix with $|\mathcal{S}||T|$ cells, for the set of sources $\mathcal{S}$ and timestamps T.

Finally, both global and dynamic responsibility computations are embarrassingly parallel across users, enabling straightforward parallelisation [26].

5.3 Implementation Benchmarks

We benchmark the efficiency and correctness of three implementations of the Shapley computation:

1. A brute-force implementation of the Shapley equation given in Eq. 31.

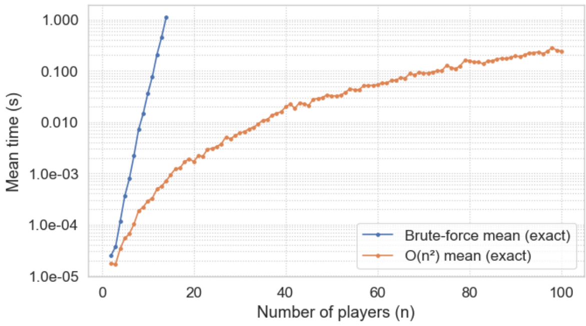


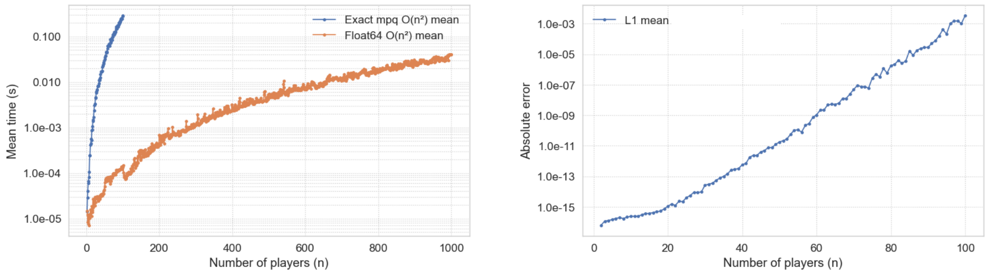
Fig. 4 Runtime comparison of brute-force and ERA implementations of Algorithm 1.

2. An exact rational arithmetic implementation of Algorithm 1 (ERA), using the `mpq` rational type to return exact fractional Shapley values.
3. An approximating floating-point implementation of Algorithm 1 (FP), accelerated using the `njit` compiler.

We first benchmark and validate the ERA implementation. Figure 4 compares the runtime of implementations (1) and (2). The x-axis denotes the number of players n in the test instance. Each test instance consists of a weight vector of length n, constructed by sampling n real numbers uniformly in $[0, 1]$ to eight decimal places and converting them to `mpq` rationals. The y-axis shows the mean computation time (log scale) for evaluating the Shapley vector. Each data point is the average over 100 independent trials. The ERA implementation computes exact Shapley values for $n = 100$ players in approximately 0.25 seconds on average. Correctness was verified by checking equality of outputs between the brute-force and ERA implementations for every trial. All experiments were conducted on a Windows 11 workstation (Intel Core Ultra 7 165H, 32GB RAM) using Python 3.12.

We next benchmark the floating-point implementation. Figure 5(a) compares the runtime of the FP implementation against ERA, using the same experimental protocol. The FP implementation computes Shapley values for $n = 1000$ players in under 0.1 seconds on average. To assess numerical accuracy, we computed the absolute error between FP and exact ERA outputs. Figure 5(b) plots the maximum absolute deviation as a function of n. The observed error grows approximately linearly with the number of players. This behaviour is expected, as floating-point evaluation of elementary symmetric polynomials involves accumulation of progressively smaller multiplicative terms in the probabilistic range.

In summary, the ERA implementation provides exact results with quadratic scaling and is appropriate when exact arithmetic is required or problem sizes are moderate. The floating point implementation offers substantial speed improvements and scales to larger instances, at the cost of controlled numerical approximation.



(a) Runtime comparison **(b)** Absolute error

Fig. 5 Runtime comparison of exact and floating-point implementations of Algorithm 1, with corresponding absolute error.

6 Empirical Study

In this section we illustrate the practical effectiveness and efficiency of the proposed responsibility measure on a real diffusion dataset study, using our ERA implementation. By *effectiveness* we mean the ability to identify the set of sources which are significantly responsible for the spread, given our assumptions. Here, a source is treated as significant if it is assigned at least 1% of the total expected responsibility. By *efficiency* we mean the empirical runtime required to compute exact Shapley attributions at the scale of a real diffusion dataset.

Section 6.1 describes the data we use and the preprocessing steps required to construct the exposure model and weights. Section 6.3 provides preliminary descriptive analysis to characterise scale, weight noise, and participation heterogeneity.

Finally, Section 6.4 reports the evaluation results, organised around our selected research questions that exploit our methods (concerning rankings, structure, temporal dynamics, baseline comparison, and runtime).

6.1 Data Source and Preparation

We first describe our data source. Our analysis is based on the *FibVID* corpus [5, 6], which records the diffusion of fact-checked news during the COVID-19 period, using anonymised Twitter data collected between February and December 2020. The dataset links fake news claims verified by *PolitiFact* and *Snopes* to online sources. In total, FibVID contains 1,353 verified claims and 221,253 associated tweets generated by 144,741 unique users. Tweet records include identifiers, timestamps, retweet relationships, sources, hashtags, and user metadata. For each tweet–claim pair, a word2vec cosine similarity score in $[0, 1]$ is provided as a graded indicator of semantic correspondence which indicates how similar the source tweet is to a given fake news claim. All users in the dataset are anonymised. For further privacy, we do not analyse or reproduce the textual content of individual tweets in this paper. A detailed description of the corpus and its collection process is provided in the original study [5].

6.2 Data Preparation

To focus on the early stages of diffusion for the purpose of root–cause analysis, we restrict attention to posts labelled as COVID–Fake within the period February–July 2020, prior to the saturation observed later in the dataset. From this subset, we extract a CSV representation of the data, the schema of which is given in Table 1.

Source–user pairs are constructed by identifying each source in the original dataset and defining its recipients as the set of descendants in the corresponding retweet cascade. For the purposes of this study, it is assumed that if a misinformation tweet was reposted, this is evidence for being misinformed. Accordingly, if an original tweet has a certain probability of being misinformation, a re-poster has the same corresponding probability of being misinformed.

Weights are derived from word2vec similarity scores between posts and fact-checked fake-news claims. We treat these scores as noisy proxies for the probability that a post is misinformation, and use them directly as parameters in the noisy–OR exposure model. They are not interpreted as calibrated probabilities, but suggest a common monotone scale on which relative causal influence can be approximated. The empirical study is therefore best viewed as a proof of concept illustrating the performance of the method at a given scale and data distribution. We emphasise that any results about responsibility are therefore expected to improve in light of developing a high-performing and calibrated misinformation detector, which is out of scope for this paper.

Overall, the processed dataset contains the information required to instantiate a noisy–exposure model of the form illustrated in Figure 2.

Table 1 Data-schema for the prepared dataset.

Field	Type	Description
`source`	Integer (ID)	Identifier of the original poster of a COVID–19 fake-news tweet.
`user`	Integer (ID)	Identifier of a downstream user who reposted somewhere in a chain of reposts.
`timestamp`	Datetime (UTC)	Time at which the tweet was reposted or observed.
`tweet_id`	Integer (ID)	Identifier of the original tweet being propagated.
`weight`	Float $[0, 1]$	Semantic similarity score between the tweet and a reference fake-news claim, interpreted as an estimated probability of misinformation.

6.3 Data Analysis

The purpose of this section is to give the reader a high-level feel for the dataset and to explain why its structural properties make it a suitable setting for our method. In particular, the data exhibits three features that our model is designed to address: (i) noisy misinformation signals, (ii) overlapping exposures in which users encounter multiple related posts from different sources.

We now present high-level statistics of the prepared dataset in Tables 2 and 3. Table 2 reports the overall scale and temporal coverage, including the number of tweets, users, distinct posts, sources, and the observation window. Table 3 summarises distributional statistics for four key quantities: (i) reposts per user, capturing user-level participation; (ii) re-posters per post, capturing propagation breadth; (iii) posts per day, capturing temporal intensity; and (iv) weight scores, representing estimated misinformation propensities.

Table 2 Scale and coverage statistics of the dataset

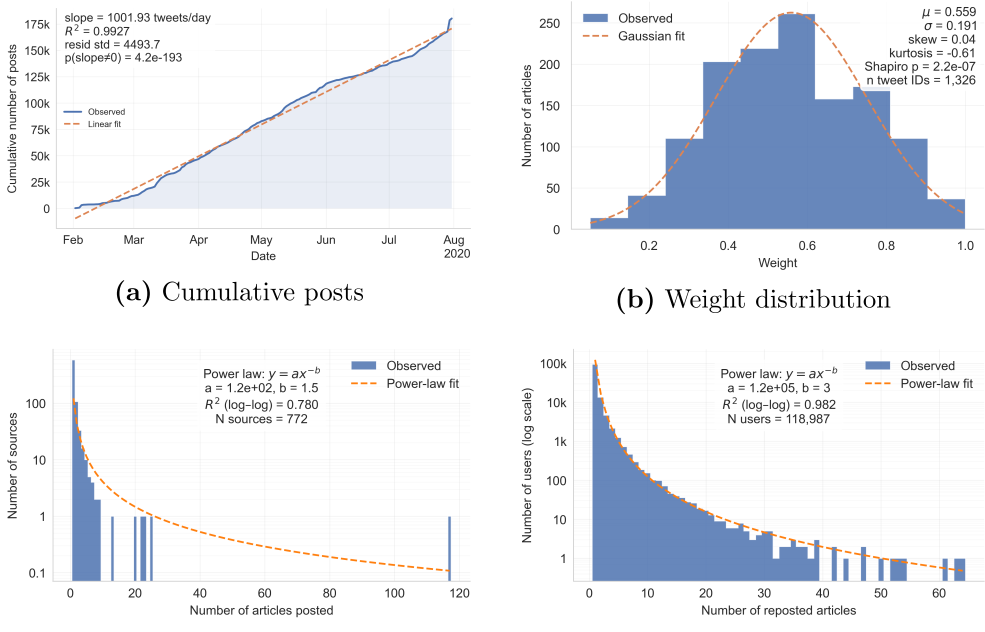
Statistic	Value	Description
Scale and coverage		
Posts	180,303	Total number of tweets (one per row).
Users	118,987	Number of different users who reposted.
posts	1,326	Number of distinct `tweet_id`s posted by sources.
Sources	772	Number of different sources who posted a post.
$\mathbf{v}(A)$	68,773.78	Expected spread $\mathbf{v}(A)$
Start date	2020-02-02	Date of the earliest tweet.
End date	2020-07-31	Date of the latest tweet.
Time span (days)	181	Length of the observation window.

Table 3 High-level distributional statistics of the dataset

Statistic	Reposts per user	Re-posters per post	posts per day	Weight
Mean	1.46	131.2	996.1	0.559
Median	1.0	63.0	875.0	0.553
Mode	1.0	4.0	1,193	1.0
Std	1.58	273.9	932.0	0.191
Min	1.0	1.0	9.0	0.051
Max	64.0	6,947	10,725	1.0

To describe the dataset in more detail, we present four graphs in Figure 6, which describe key dynamics and distributions of the dataset. (a) shows that the cumulative number of tweets grows approximately linearly over time, indicating a stable rate of activity rather than episodic bursts. (b) shows that the distribution of weights is approximately normal, reflecting the presence of noise in the misinformation signals. (c) is a histogram of the number of posts sources post. In general, this is observed to approximately follow a power law distribution – a few key sources post a lot more than others; most sources post one post, and one anomalous outlier posts 150+ posts. Finally, (d) is a histogram describing the number of unique posts per user, which describes the number of posts users repost. The distribution follows a heavy-tailed (approximately power-law) form, indicating strong heterogeneity in user participation, with many low-activity users and a small number of highly active users.

Taken together, these patterns reflect the characteristic combination of linear growth, measurement noise, and over-exposure effects which we might typically observe in many similarly-scaled social-media diffusion processes. This combination is consistent with the assumptions under which our method is designed to operate (i.e. noise and over-exposure), and thus supports the appropriateness of the dataset for our study.



(a) Cumulative posts

(b) Weight distribution

(c) Posts per source

(d) Re-posts per user

Fig. 6 Overview of dataset dynamics and distributions.

Note. Our aim is to illustrate and compare the proposed method on this dataset, rather than to provide a comprehensive data analysis of the dataset itself. For a fuller account of the FibVID dataset, see the original dataset paper [5].

6.4 Evaluation

In this section we present the empirical results and organise the analysis around five questions:

1. *Who were the significantly responsible sources, and how responsible were they?*
2. *How are the significantly responsible sources co-operating?*
3. *How does responsibility evolve over time for the most significant sources?*
4. *Does the responsibility measure rank sources differently from the baseline measures?*
5. *How efficient is it to compute responsibility in practice?*

Table 4 Top-10 sources ranked by c_{resp} (% of total spread), with cross-measure ranks and scores. Final row shows computation time (seconds) for each measure.

user ID	$c_{\text{resp}}(\%)$	r_{resp}	c_{resp}	r_{suff}	c_{suff}	r_{nec}	c_{nec}	r_{sum}	c_{sum}
51591	21.21	1	14,589.05	1	17,128.59	1	12,558.27	1	20,404.41
27011	2.70	2	1,854.73	2	2,466.41	2	1,510.18	2	2,594.87
143876	1.62	3	1,115.59	4	1,508.29	3	900.74	4	1,558.00
93248	1.60	4	1,102.00	3	1,550.25	5	873.48	3	1,614.36
143737	1.41	5	968.24	6	1,091.15	4	890.05	6	1,125.40
9225	1.29	6	884.88	5	1,193.99	6	718.33	5	1,281.56
95758	1.21	7	830.33	8	1,029.23	7	708.01	8	1,055.01
92676	1.12	8	772.17	7	1,076.66	8	614.61	7	1,123.58
141002	0.95	9	651.29	10	850.92	9	541.31	10	864.42
60201	0.92	10	630.43	9	908.34	14	485.28	9	964.14
Time (s)	–	–	**7.76**	–	**2.82**	–	**3.17**	–	**1.70**

The first three questions illustrate the substantive use of the proposed measure for analysing a real diffusion process. The final two questions assess methodological properties: whether the Shapley-based measure yields materially different conclusions from simpler baselines, and whether the computation is feasible at the scale of a realistic dataset.

Q1: Who were the significantly responsible sources, and how responsible were they?

Table 4 reports the ten sources with the highest responsibility, ranked by $c_{\text{resp}}(\%)$, the proportion of total expected spread attributed to each source. Each row corresponds to one anonymised source. For every measure the table shows a *rank–score* pair: $r_\bullet$ denotes the rank (smaller is better) and $c_\bullet$ the raw score. Thus $(r_{\text{resp}}, c_{\text{resp}})$ corresponds to Shapley responsibility, $(r_{\text{suff}}, c_{\text{suff}})$ to the sufficiency baseline, $(r_{\text{nec}}, c_{\text{nec}})$ to the necessity baseline, and $(r_{\text{sum}}, c_{\text{sum}})$ to the sum-of-weights baseline. Here, c_{sum} is simply the sum of a source's outgoing weights in the Noisy–OR exposure model, and estimates the number of misinformation posts they exposed recipients to. The final row reports wall–clock computation time for each measure on the same graph instance.

We call a source *significantly responsible* when $c_{\text{resp}}(\%) \geq 1\%$. Under this threshold the top eight rows constitute the significant set. Responsibility is highly concentrated: source 51591 alone accounts for 21.21% of the total expected spread—over an order of magnitude larger than any other actor. The next seven significant sources each contribute between roughly 1% and 3%, while all remaining sources fall below the threshold. This indicates that the diffusion is driven by a small core of accounts rather than by a diffuse long tail, providing a concrete answer to the motivating question of which actors were most responsible.

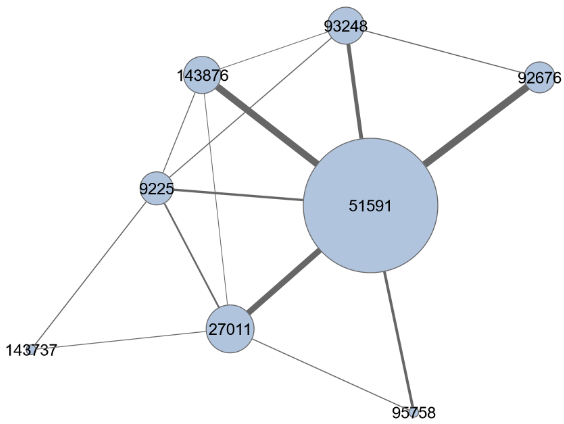


Fig. 7 **Operational network of significantly responsible sources.**

Q2: How are the significantly responsible sources co-operating?

Figure 7 shows the *operational network* induced by the significantly responsible sources ($c_{\text{resp}}(\%) \geq 1\%$). Each such source appears as a node whose size is proportional to its responsibility score. An undirected edge is drawn whenever one source has reposted another at least once; the edge width equals the expected number of misinformation items reposted in either direction, capturing the strength of their direct relationship.

The resulting structure provides a qualitative map of the core misinformation ecosystem. Several of the high-responsibility accounts are connected through reposting behaviour, and the most responsible source occupies a central position in this subgraph. This suggests that responsibility is not merely the sum of isolated actors but is supported by a network of mutually reinforcing exposures.

Q3: How does responsibility evolve over time for the significantly responsible sources?

Figures 8 and 9, derived from the dynamic responsibility matrix, track how responsibility accumulates during the diffusion process. In both plots each significantly responsible source is shown as a distinct colour, while all others are aggregated into an "Others" category.

Figure 8 displays *absolute responsibility* over time. The y-axis equals the expected number of users exposed to misinformation (the value of $\mathbf{v}$ at that time), so the final stack height of 68,773.78 matches the total expected spread. At this resolution (with the start of the spread obscured due to scale), there appears to be a roughly-linear growth of the major bands indicates that the leading sources accumulate responsibility steadily rather than through isolated bursts.

Figure 9 normalises the same quantities to percentages. This reveals relative shifts, particularly at the start of the spread: the leading source reaches roughly 10% responsibility within the first half-month almost immediately, with the responsibility distributions stabilising with scale. Such views allow analysts to identify when new

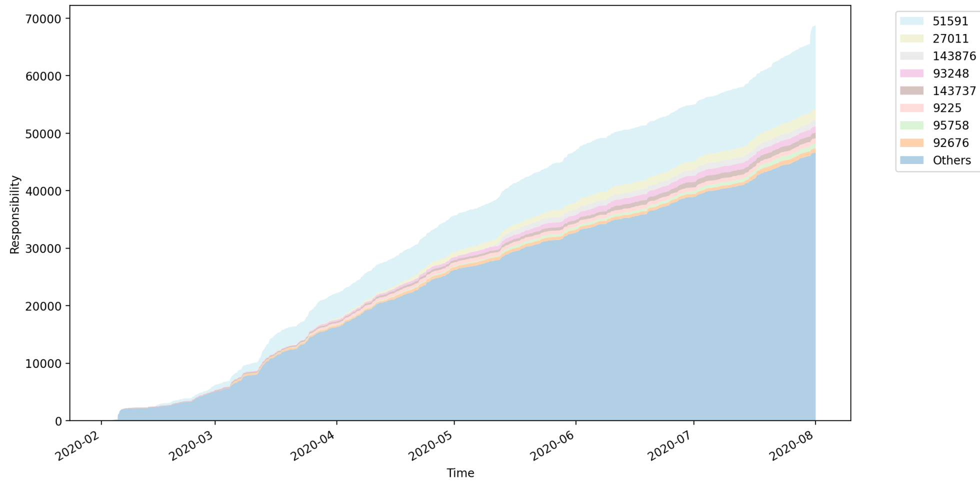


Fig. 8 Raw responsibility over time

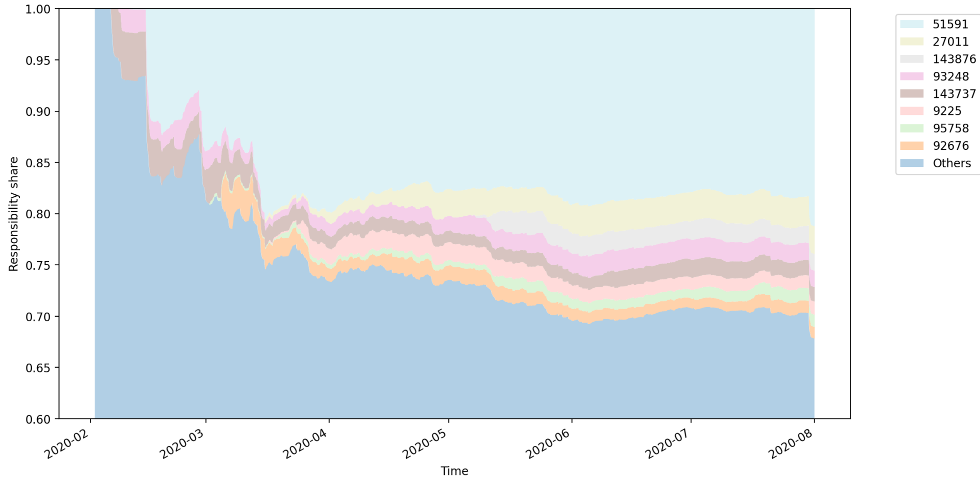


Fig. 9 Proportion of responsibility over time

actors become operationally important, and when responsibility proportions change. Activity corresponding to changepoints in the figure can be investigated as potentially significant events which affect the discourse.

The temporal analysis also provides an empirical stability check on the responsibility rankings. Although the rankings are initially volatile, the leading sources stabilise after the early phase of the cascade and remain largely unchanged over the subsequent observation period. In this dataset, the principal sources identified after the volatile period are therefore not artefacts of the final aggregation window: they persist as the

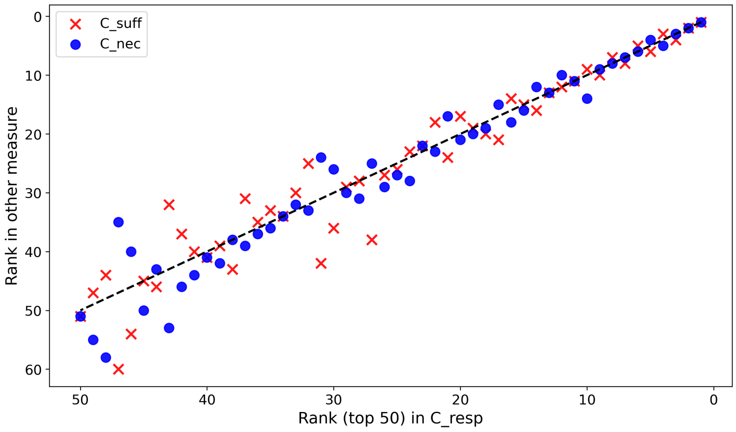


Fig. 10 **Rank agreement against baseline measures.**

diffusion process unfolds. This supports the interpretation that the high-responsibility sources reflect stable structural features of the observed cascade, rather than transient fluctuations at the end of the data collection period.

Q4: Does the responsibility measure rank sources differently from the baseline measures?

Figure 10 compares the ranking induced by C_{resp} with those from C_{suff} and C_{nec} for the top-50 sources (the C_{sum} baseline was omitted because it was empirically ranking-equivalent to C_{suff}). Each point shows a source's rank under C_{resp} on the x-axis and its rank under the comparison measure on the y-axis. Points on the diagonal indicate identical rankings, while vertical displacement indicates disagreement.

The scatter shows broad agreement for the most influential accounts but increasing divergence further down the ranking. To quantify this, we computed Spearman rank correlations over the top-50 sources. Correlations between C_{resp} and the baselines lie in the range 0.96–0.98, indicating strong but not perfect concordance. Thus the Shapley-based measure produces materially different orderings while remaining broadly corroborative with the other measures on this dataset. This close corroboration might be explained by the fact given all three measures are theoretically related; insofar as they are all expressed in terms of our global exposure function $\mathbf{v}$, itself defined in terms of a Noisy–OR model.

Q5: How efficient is it to compute responsibility in practice?

The final row of Table 4 reports computation time for each measure on the full dataset. The Shapley computation required 7.76 seconds, compared with 1.70–3.17 seconds for the baselines. More demanding is the construction of the dynamic responsibility matrix used for Figures 8 and 9, which required approximately four minutes (3:53–4:03 across repeat runs). Given that the method yields *exact* Shapley attributions for every user and time point, we think the observed runtimes are practically acceptable for forensic analysis of misinformation spreads.

7 Discussion

Primary use-cases.

The proposed responsibility methods are intended for two practical settings: (i) *forensic root–cause analysis* of information spreads over a narrow well-defined topic or specific piece of information, and (ii) *dynamic monitoring* of responsibility as a spread unfolds. Because the Shapley formulation satisfies efficiency, individual contributions sum exactly to the total expected spread, allowing responsibility to be *stacked* across time, sources, or subpopulations without double counting. This property is particularly valuable when analysts must explain how much of the observed diffusion is attributable to specific actors or groups. Consequently, statements of the form "user X was $K\%$ responsible for the spread" can be given a principled interpretation via the measure $C_{\mathrm{resp}\%}$, and at any point in the evolution of the spread.

Generality of the model.

Although we describe weights in terms of information exposure, the framework is not limited to that domain. The same construction applies to any process that can be expressed as noisy influence on a binary outcome—e.g., disinformation, coordinated advertising, radicalisation content, or diffusion of rumours or scientific claims. Likewise, the outcome need not be mere exposure: the weights could represent approval, belief change, or any graded response for which a probabilistic propagation model is available.

Complementarity with existing metrics.

The method is not proposed as a replacement for other measures or simpler heuristics such as C_{suff} or C_{nec}. Rather, our Shapley based responsibility measure provides a principled attribution that accounts for interaction effects between sources, revealing cases where influence is jointly created rather than individually obvious. In practice we expect investigators to use a portfolio of measures, where our responsibility methods can serve as a reference when a principled, interpretable and precise attribution is required.

Assumptions and limitations.

The attribution provided by C_{resp} is conditional on the noisy–OR exposure model and on the quality of the weight estimates. If the weights systematically over- or underestimate the probability that a post will misinform a user, the absolute magnitudes of responsibility will be affected, although the relative ordering may remain stable. The model also assumes independence of exposures given the weights; coordinated or adversarial behaviour that violates this assumption could require richer models. Dependencies between exposures would break the multiplicative structure of the current noisy–OR model and require a different attribution framework, which does not automatically promise a comparably efficient solution and which we leave to future work. Finally, responsibility as defined here measures a notion of *causal responsibility*, not intent, blame, harm, or legal liability, and should be interpreted accordingly.

An additional modelling choice concerns the locus of attribution. Responsibility is assigned exclusively to root sources—i.e., to the original source–post IDs that initiate exposure—and not to intermediary information brokers who relay or amplify content. The resulting measure therefore best captures a notion of *ultimate causal responsibility* for the spread among content originators, rather than transmission responsibility among intermediaries. Alternative attribution schemes that distribute responsibility across diffusion paths are possible, but would require a different formalisation, and so is also left to future work.

Finally, we discuss limitations with our empirical study. We emphasise that the study is intended as a proof of concept for the proposed responsibility measure, rather than as a full misinformation-detection pipeline. The word2vec similarity scores are used because they are provided with the FibVID dataset and give a reproducible, topic-relevant weighting scheme for constructing a Noisy–OR exposure model. In operational applications, these weights should ideally be supplied by a calibrated misinformation detector. Developing such a detector is outside the scope of this paper: our contribution is the responsibility-allocation method, not the upstream classification of misinformation content.

8 Conclusion

We developed a principled framework for attributing expected responsibility in information diffusions under noisy and overlapping exposures. By formalising user-level propagation as a cooperative Noisy–OR game, we cast responsibility allocation as a well-posed cooperative value problem and identified the Shapley value as the unique rule satisfying efficiency, symmetry, null-player, and additivity requirements. This establishes a formally grounded notion of *expected responsibility* in probabilistic diffusion settings.

Beyond the conceptual contribution, we derived a closed-form characterisation of the Shapley value for Noisy–OR games and obtained an exact $O(n^2)$ algorithm for its computation. This removes the need for sampling-based approximations and makes exact attribution computationally feasible for realistic cascades, including dynamic settings.

Empirical evaluation on the FibVID dataset suggests that responsibility is highly concentrated, that significantly responsible sources form a partially connected operational core, and that the Shapley-based measure produces materially different rankings from simpler heuristics. Exact dynamic attribution over the full dataset was computable within minutes, confirming practical tractability.

More broadly, this work provides a formal bridge between cooperative game theory and empirical diffusion analysis. The combination of axiomatic grounding, exact computation, and empirical scalability distinguishes the framework from heuristic measures of social influence. It enables precise, quantitative, and interpretable answers to questions of the form "who was responsible for the spread, by how much, and at what stage?", while preserving principled axiomatic guarantees. We propose this framework as a rigorous analytic tool for studying causal responsibility in online information ecosystems.

Future work.

Several directions follow directly from this study. First, the framework could be applied with alternative sources of weights, such as human fact-check confidence scores or classifier posteriors, enabling comparison across detection pipelines. Second, propagating uncertainty in the weights through the Shapley computation would allow interval estimates of responsibility rather than point values. Third, cross-platform studies could compare the patterns of responsibility across topics and networks.

Declarations

- **Acknowledgements**. A preliminary version of this work was developed for The Defence Science and Technology Laboratory (Dstl), under contract with Frazer-Nash Consultancy Ltd.
- **Funding**. The work was conducted by the author as part of their employment at Frazer-Nash Consultancy Ltd. No external funding was received for this work.

- **Availability of data and materials**. The dataset supporting the conclusions of this article is available in the GitHub repository: https://github.com/anonymous-expected-responsibility/paper. The repository also contains the source code and Jupyter notebooks required to reproduce all experiments presented in this article.
- **Author contributions**. The sole author conceived the study, developed the methodology, implemented the experiments, analysed the results, and wrote the manuscript.
- **Competing interests**. The author declares no other competing interests.
- **Supplementary Information**. The Appendix contains the supporting proofs and pseudo-code underlying the closed-form Noisy–OR Shapley computation.

References

[1] Rozemberczki, B., Watson, L., Bayer, P., Yang, H.-T., Kiss, O., Nilsson, S., & Sarkar, R. *The Shapley Value in Machine Learning*. Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence and the Twenty-Fifth European Conference on Artificial Intelligence (IJCAI–ECAI 2022). International Joint Conferences on Artificial Intelligence Organization, 2022. Available at: https://www.ijcai.org/proceedings/2022/778 (accessed 2026).

[2] PolitiFact. *PolitiFact: The Truth-O-Meter*. Poynter Institute, 2007–present. Available at: https://www.politifact.com (accessed 2026).

[3] Snopes. *Snopes: Urban Legends, Rumours, and Misinformation*. Snopes Media Group, 1994–present. Available at: https://www.snopes.com (accessed 2026).

[4] Savage, C. D. *Symmetric Functions and Combinatorial Identities*. University Lecture Notes, 2023.

[5] Kim, J., Jisu, K., et al. FibVID: comprehensive fake news diffusion dataset during the COVID-19 period. *Telematics and Informatics*, 64:101688, 2021.

[6] merry555. *FibVID GitHub Repository.* Available at: https://github.com/merry555/FibVID, accessed 2026.

[7] Shapley, L. S. A value for n-person games. In *Classics in Game Theory*, pp. 69–79. Princeton University Press, 2020.

[8] Tokita, C. K., Aslett, K., Godel, W. P., Sanderson, Z., Tucker, J. A., Nagler, J., Persily, N., & Bonneau, R. *Measuring receptivity to misinformation at scale on a social media platform. PNAS Nexus*, 3(10), pgae396, 2024.

[9] Mosleh, M., & Rand, D. G. *Measuring exposure to misinformation from political elites on Twitter. Nature Communications*, 13, 7144, 2022.

[10] Avram, M., Micallef, N., Patil, S., & Menczer, F. *Exposure to social engagement metrics increases vulnerability to misinformation. HKS Misinformation Review*, 1(5), 2020.

[11] Del Vicario, M., Bessi, A., Zollo, F., Petroni, F., Scala, A., Caldarelli, G., Stanley, H. E., & Quattrociocchi, W. *The spreading of misinformation online. Proceedings of the National Academy of Sciences*, 113(3), 554–559, 2016.

[12] Pierri, F., Piccardi, C., & Ceri, S. *Topology comparison of Twitter diffusion networks effectively reveals misleading information. Scientific Reports*, 10(1), 1372, 2020.

[13] Kempe, D., Kleinberg, J., & Tardos, É. *Maximizing the spread of influence through a social network. Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 137–146, 2003.

[14] Budak, C., Agrawal, D., & El Abbadi, A. *Limiting the spread of misinformation in social networks. Proceedings of the 20th International Conference on World Wide Web*, 665–674, 2011.

[15] Narayanam, R., & Narahari, Y. *A Shapley value-based approach to discover influential nodes in social networks. IEEE Transactions on Automation Science and Engineering*, 8(1), 130–147, 2011.

[16] Papapetrou, P., et al. *A Shapley-value approach for influence attribution. Proceedings of the 15th Pacific-Asia Conference on Advances in Knowledge Discovery and Data Mining (PAKDD)*, 138–149, 2011.

[17] Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., et al. *The science of fake news. Science*, 359(6380), 1094–1096, 2018.

[18] Granovetter, M. *Threshold models of collective behavior. American Journal of Sociology*, 83(6), 1420–1443, 1978.

[19] Chen, W., Wang, C., & Wang, Y. *Scalable influence maximization for prevalent viral marketing in large-scale social networks. Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1029–1038, 2010.

[20] Vosoughi, S., Roy, D., & Aral, S. *The spread of true and false news online. Science*, 359(6380), 1146–1151, 2018.

[21] Lundberg, S. M., & Lee, S.-I. *A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems*, 30, 2017.

[22] Chalkiadakis, G., Elkind, E., & Wooldridge, M. *Computational aspects of cooperative game theory.* Morgan & Claypool Publishers, 2011.

[23] Judea Pearl. *Causality: Models, Reasoning, and Inference.* Cambridge University Press, 2nd edition, 2009.

[24] Halpern, J. Y., & Pearl, J. *Causes and explanations: A structural-model approach. Part I: Causes. The British Journal for the Philosophy of Science*, 56(4), 843–887, 2005.

[25] Chockler, H., & Halpern, J. Y. *Responsibility and blame: A structural-model approach. Journal of Artificial Intelligence Research*, 22, 93–115, 2004.

[26] Ian Foster. *Designing and Building Parallel Programs.* Addison–Wesley, 1995. Available at: https://www.mcs.anl.gov/~itf/dbpp/ (accessed 2026).

A Proofs

In this appendix we present the proofs and pseudo-code underlying our implementation.

A.1 Main notation

For ease of reference, we summarise the principal notation used throughout the proofs and the paper.

Exposure model.

- V — set of users.
- T — set of timestamps.
- $\mathcal{A}$ — set of source-post IDs (posts).
- $E \subseteq V \times V \times T$ — exposure events (s, r, t).
- $\rho : E \to \mathcal{A}$ — mapping from exposure events to source-post IDs.

- $\mathcal{S} \subseteq V$ — set of sources.
- $\mathrm{art}(s) \subseteq \mathcal{A}$ — posts authored by source s.

Noisy–OR exposure model.

- $w_i \in (0, 1]$ — causal propensity of post i to misinform.
- $a_i := 1 - w_i$ — complement probability.
- $v_u(A)$ — probability that user u is misinformed by posts A.
- $\mathbf{v}(A) := \sum_{u \in V} v_u(A)$ — expected spread generated by A.

Cooperative games and responsibility.

- N — set of players (posts) in a game.
- $v : 2^N \to \mathbb{R}_{\geq 0}$ — value function.
- $\phi_i(N, v)$ — Shapley value of post i.
- $C_{\mathrm{resp}}(s)$ — expected responsibility of source s.
- $C_{\mathrm{resp}\%}(s)$ — percentage expected responsibility of source s.

Dynamic responsibility.

- $t_1 < \cdots < t_T$ — ordered timestamps.
- $\mathbf{R} \in \mathbb{R}^{|\mathcal{S}| \times T}$ — dynamic responsibility matrix.
- $\mathbf{R}_{s,t}$ — incremental responsibility of source s at time t.

Algorithmic notation.

- n — number of posts in a Noisy–OR game.
- n_u — number of posts exposed to user u.
- e_k, e_k^{-i} — elementary symmetric polynomials (defined in Appendix A).

It will also help to have the following notation about symmetric polynomials in order to simplify the presentation and proofs.

Definition 10 (Elementary symmetric polynomials over N) let $N := [n]$ be a set of indices, and let $a_1, \ldots, a_n$ be real numbers corresponding to each index. For convenience, we abbreviate $N_m = [m]$ when $1 \leq m \leq |N|$ (i.e. first m elements of N).

For $S \subseteq N$, $0 \leq k \leq |N|$, we have the following:

$$\mathcal{P}_k(S) \coloneqq \{ C \subseteq S \mid |C| = k \} \qquad \text{(S's k-sized subsets)} \tag{12}$$

$$e_k(S) \coloneqq \sum_{T \in \mathcal{P}_k(S)} \prod_{j \in T} a_j \qquad \text{(k-th elementary symmetric polynomial over S)} \tag{13}$$

$$e_k \coloneqq e_k(N) \qquad \text{(full symmetric polynomial on N)} \tag{14}$$

$$e_k^m \coloneqq e_k(N_m) \qquad \text{(partial symmetric polynomial on N_m)} \tag{15}$$

$$e_k^{-i} \coloneqq e_k\big(N \setminus \{i\}\big) \qquad \text{(leave-one-out symmetric polynomial on N)} \tag{16}$$

By convention $e_0(S) = 1$. For any $S \subseteq N$, the quantity $e_k(S)$ defined in Eq. (13) is the standard k-th elementary symmetric polynomial in the variables $(a_j)_{j \in S}$. Our notation follows the conventions of Savage [4], in particular:

- e_k (Eq. (14)) is the elementary symmetric polynomial over $(a_j)_{j \in N}$,
- e_k^m (Eq. (15)) is the elementary symmetric polynomial over $(a_j)_{j \in N_m}$,
- e_k^{-i} (Eq. (16)) is the elementary symmetric polynomial over $(a_j)_{j \in N \setminus \{i\}}$.

Example 4 We illustrate definition 10 with a small example. Let $N = \{1, 2, 3\}$ with associated values a_1, a_2, a_3.

Subsets $\mathcal{P}_k(S)$ (Eq. (12)). For $S = N$ we have

$$\begin{aligned}
\mathcal{P}_0(N) &= \{\varnothing\}, \\
\mathcal{P}_1(N) &= \{\{1\}, \{2\}, \{3\}\}, \\
\mathcal{P}_2(N) &= \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}, \\
\mathcal{P}_3(N) &= \{\{1, 2, 3\}\}.
\end{aligned}$$

Elementary polynomials $e_k(S)$ (Eq. (13)). Using (13), for $S = N$ we obtain

$$\begin{aligned}
e_0(N) &= 1, \\
e_1(N) &= a_1 + a_2 + a_3, \\
e_2(N) &= a_1 a_2 + a_1 a_3 + a_2 a_3, \\
e_3(N) &= a_1 a_2 a_3.
\end{aligned}$$

Given Eq. (14), we also have the notation $e_k = e_k(N)$, for each $k \in \{0\} \cup N$.

Partial symmetric polynomials e_k^m (Eq. (15)). Using (15),

$$e_1^2 = e_1(N_2) = a_1 + a_2, \qquad e_2^2 = e_2(N_2) = a_1 a_2.$$

Leave-one-out symmetric polynomials e_k^{-i} (Eq. (16)). For $i = 1$, we have

$$e_1^{-1} = a_2 + a_3, \qquad e_2^{-1} = a_2 a_3.$$

A.2 Proof – Marginal Contribution in the Noisy–OR game

Proposition 1 (Marginal contribution in Noisy–OR game) *Let (N, v) be a Noisy–OR game. For any player $i \in N$ and any coalition $S \subseteq N \setminus \{i\}$,*

$$\boxed{v(S \cup \{i\}) - v(S) \;=\; w_i \prod_{j \in S} a_j} \tag{17}$$

Proof Let (N, v) be a Noisy–OR game, $i \in N$ be a player and $S \subseteq N \setminus \{i\}$ a coalition. We begin with the following setup: By Def. 3 of the Noisy–OR game:

$$v(S) \;=\; 1 - \prod_{j \in S} (1 - w_j) \tag{18}$$

where $w_j \in [0,1]$ is the success probability of player j and $a_j := 1 - w_j$ its complement. For clarity of presentation we use the following abbreviation:

$$P_S \;:=\; \prod_{j \in S} (1 - w_j) \tag{19}$$

Consequently, $v(S) \;=\; 1 - P_S$ (by substitution). The proof is completed as follows:

$$
\begin{aligned}
v(S \cup \{i\}) &= 1 - \prod_{j \in S \cup \{i\}} (1 - w_j) && \text{(by Def. 3)} && (20)\\
\ldots &= 1 - (1 - w_i) \prod_{j \in S} (1 - w_j) && \text{(expansion given } i \notin S\text{)} && (21)\\
\ldots &= 1 - (1 - w_i) P_S && \text{(by abbreviation of } P_S\text{)} && (22)\\
\ldots &= 1 - \big(P_S - w_i P_S\big) && \text{(expanding } (1 - w_i)P_S\text{)} && (23)\\
\ldots &= 1 - P_S + w_i P_S && \text{(distributing the outer minus sign)} && (24)\\
\ldots &= v(S) + w_i P_S && \text{(as } v(S) = 1 - P_S\text{)} && (25)\\
\ldots &= v(S) + w_i \prod_{j \in S} (1 - w_j) && \text{(by abbreviation of } P_S\text{)} && (26)\\
v(S \cup \{i\}) &= v(S) + w_i \prod_{j \in S} a_j && \text{(since } a_j = 1 - w_j\text{)} && (27)\\
v(S \cup \{i\}) - v(S) &= w_i \prod_{j \in S} a_j && \text{(subtracting } v(S) \text{ from both sides)} && (28)
\end{aligned}
$$

This completes the proof of Equation (17). □

A.3 Proof – Shapley Equation for the Noisy–OR game

Proposition 2 (Equation for the Noisy–OR game) *Let $N := [n]$ be a set of players, and let (N, v) be a Noisy–OR game. Then the Shapley value of any player $i \in N$ satisfies*

$$\boxed{\phi_i \;=\; \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{e_k^{-i}}{\binom{n-1}{k}}} \tag{29}$$

where e_k^{-i} denotes the k-th elementary symmetric polynomial of the vector $(a_j)_{j \in [n] \setminus \{i\}}$.

Proof Let $N := [n]$ and (N, v) be a Noisy–OR game. Recall Def. 4 of the Shapley value:

$$\phi_i \;=\; \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{\binom{n-1}{k}} \sum_{S \in \mathcal{P}_k(N \setminus \{i\})} \big(v(S \cup \{i\}) - v(S)\big). \tag{30}$$

Recall Proposition 1: For any $S \subseteq N \setminus \{i\}$, $v(S \cup \{i\}) - v(S) \;=\; w_i \prod_{j \in S} a_j$. By substitution we have:

$$\phi_i \;=\; \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{\binom{n-1}{k}} \sum_{S \in \mathcal{P}_k(N \setminus \{i\})} \Big(w_i \prod_{j \in S} a_j\Big). \tag{31}$$

Since w_i does not depend on S or k, we can factor it out (using distributivity of multiplication):

$$\phi_i \;=\; \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{1}{\binom{n-1}{k}} \sum_{S \in \mathcal{P}_k(N \setminus \{i\})} \prod_{j \in S} a_j. \tag{32}$$

Recall Definition 10 of leave-one-out symmetric polynomials: $e_k^{(-i)} = \sum_{S \in \mathcal{P}_k(N \setminus \{i\})} \prod_{j \in S} a_j$, defined over a given S and N. Substituting this into Eq. (32) gives

$$\phi_i \;=\; \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{1}{\binom{n-1}{k}}\, e_k^{(-i)} \tag{33}$$

distributing over the sum, then multiplying gives:

$$\phi_i \;=\; \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{e_k^{-i}}{\binom{n-1}{k}} \tag{34}$$

□

A.4 Proof – Recurrences for Symmetric Polynomials

In this section we provide a set of recurrence equations which can be used to efficiently compute the elementary symmetric polynomials underlying the equation of Proposition 2. These equations underpin Algorithm 1.

Lemma 3 (Recurrence equations for elementary symmetric polynomials) *Let $N = [n]$, and let $a = (a_1, \ldots, a_n)$ be a vector of values each in the range $[0, 1]$. For $S \subseteq N$ and $i \in N$, define*

$$\mathcal{P}_k(S) \coloneqq \{\, C \subseteq S \mid |C| = k \,\} \qquad \textit{(S's k-sized subsets)} \tag{35}$$

$$e_k(S) \coloneqq \sum_{T \in \mathcal{P}_k(S)} \prod_{j \in T} a_j \qquad \textit{(k-th elementary symmetric sum over S).} \tag{36}$$

Adopt the convention $e_t(H) = 0$ when $t < 0$ or $t > |H|$ (and $\prod_{j \in \varnothing} a_j = 1$). Then, for all $S \subseteq N$ and $i \in S$,

$$e_0(S) = 1 \tag{37}$$

$$e_k(S) = e_k\big(S \setminus \{i\}\big) + a_i\, e_{k-1}\big(S \setminus \{i\}\big), \qquad \textit{for all } k \geq 1. \tag{38}$$

Proof We identify our set–indexed sums with the classical elementary symmetric polynomials. Given a vector $a = (a_1, \ldots, a_n)$ and any index set $S \subseteq N$, define

$$e_k(S) \;\equiv\; e_k\big((a_j)_{j \in S}\big),$$

i.e. $e_k(S)$ is the kth elementary symmetric polynomial in the variables $\{a_j : j \in S\}$. In particular, $e_0(S) = 1$ by the empty–product convention. Now, the standard recurrence for elementary symmetric polynomials (see Lemma 2.4.2 in Savage [4]) states that for variables $x_1, \ldots, x_m$,

$$e_k(x_1, \ldots, x_m) \;=\; e_k(x_1, \ldots, x_{m-1}) \;+\; x_m\, e_{k-1}(x_1, \ldots, x_{m-1}) \quad (k \geq 1),$$

with $e_0(\cdot) = 1$. Apply this identity with $(x_1, \ldots, x_m) = \big(a_j\big)_{j \in S}$, after ordering S so that its last element is $i \in S$. Then $(x_1, \ldots, x_{m-1})$ corresponds to $\big(a_j\big)_{j \in S \setminus \{i\}}$, and $x_m = a_i$, yielding

$$e_k(S) \;=\; e_k\big(S \setminus \{i\}\big) \;+\; a_i\, e_{k-1}\big(S \setminus \{i\}\big) \quad (k \geq 1),$$

while $e_0(S) = 1$ holds for all S. These are exactly (37)–(38). □

Lemma 4 (Recurrences equations) *Let $N := [n]$, where $a = (a_1, \ldots, a_n)$ is a vector of values each in the range $[0, 1]$. For $S \subseteq N$, each $i \in N$, and each $0 \leq k, m \leq |N|$ we add to the definitions of Lemma 4 the following*

$$N_m := \{1, \ldots, m\},\ \textit{when}\ m > 0 \qquad \textit{(first } m \textit{ players in } N\textit{)} \tag{39}$$

$$e_k^m := e_k(N_m) \qquad \textit{(partial symmetric sum on } N_m\textit{)} \tag{40}$$

$$e_k^{-i} := e_k\big(N \setminus \{i\}\big) \qquad \textit{(leave-one-out symmetric sum)} \tag{41}$$

By convention $N_0 := \varnothing$. Given the definitions, for each vector $a = (a_1, \ldots, a_n)$, the following equations hold:

Base conditions.

$$e_0^0 = 1, \qquad e_k^0 = 0 \qquad (1 \leq k \leq |N|), \tag{42}$$

$$e_0^m = 1 \qquad e_0^{-i} = 1 \qquad (1 \leq i, m \leq |N|), \tag{43}$$

Recurrence conditions.

$$e_k^m = e_k^{m-1} + a_m\, e_{k-1}^{m-1} \qquad (1 \leq k \leq m \leq |N|), \tag{44}$$

$$e_k^{-i} = e_k^{|N|} - a_i\, e_{k-1}^{-i} \qquad (i \in N,\ 1 \leq k \leq |N|). \tag{45}$$

Proof Fix $N := [n]$, where $a = (a_1, \ldots, a_n)$ is a vector of values each in the range $[0, 1]$.

Base condition 1: $e_0^{(0)} = 1$, and $e_k^{(0)} = 0$ for $1 \leq k \leq n$.

$e_0^{(0)} = e_0(N_0)$ (by definition of partial sum). So $e_0(N_0) = e_0(\varnothing)$ (given $N_0 := \varnothing$). By Lemma 4, $e_0(S) = 1$ for all $S \subseteq N$; in particular, for $S = \varnothing$. So $e_0(\varnothing) = 1$ (applying the Lemma). So $e_0(N_0) = 1$ (given $N_0 := \varnothing$). Hence $e_0^{(0)} = 1$ (by definition of partial sum).

$e_k^{(0)} = e_k(N_0)$ (definition of partial sum). So $e_k^{(0)} = e_k(\varnothing)$ (given $N_0 := \varnothing$). Now, $e_k(\varnothing) = \sum_{T \in \mathcal{P}_k(\varnothing)} \prod_{j \in T} a_j$. (By the definition of $e_k(S)$ Eq. (36)). For $k \geq 1$ we have $\mathcal{P}_k(\varnothing) = \varnothing$ (there are no k-subsets of the empty set), hence the sum is empty and equals 0. Equivalently, this matches the convention $e_t(H) = 0$ when $t > |H|$ with $H = \varnothing$. Therefore $e_k^{(0)} = 0$ for all $1 \leq k \leq n$.

Base condition 2: $e_0^{(m)} = 1$ for $1 \leq m \leq n$ and $e_0^{(-i)} = 1$ for all $i \in N$. For each m, $e_0^{(m)} = e_0(N_m)$ (definition of partial sum). By Lemma 4, $e_0(S) = 1$ for all $S \subseteq N$; in particular, for $S = N_m$. So $e_0(N_m) = 1$ (applying the Lemma). So $e_0^{(m)} = 1$ (by definition of partial sum). Hence $e_0^{(m)} = 1$ for every $1 \leq m \leq n$.

For each $i \in N$, $e_0^{(-i)} = e_0(N \setminus \{i\})$ (by def. of leave-one-out sum). By Lemma 4, $e_0(S) = 1$ for all $S \subseteq N$; in particular, for $S = N \setminus \{i\}$. So $e_0(N \setminus \{i\}) = 1$ (applying the Lemma) So $e_0^{(-i)} = 1$ (by def. of leave-one-out sum). Therefore $e_0^{(-i)} = 1$ for all $i \in N$.

Recurrence condition 1: $e_k^m = e_k^{m-1} + a_m\, e_{k-1}^{m-1}$ for all $(1 \leq k \leq m \leq |N|)$ Let $N_m \subseteq N$ where $1 \leq m \leq |N|$. Fix a k such $(1 \leq k \leq m \leq |N|)$. By Lemma 4 we have

$$e_k(N_m) = e_k\big(N_m \setminus \{m\}\big) + a_i\, e_{k-1}\big(N_m \setminus \{m\}\big) \tag{46}$$

Given N_m is the first m indices of N in general, $N_m \setminus \{m\} = N_{m-1}$. Accordingly, we have

$$e_k(N_m) = e_k\big(N_{m-1}\big) + a_i\, e_{k-1}\big(N_{m-1}\big) \tag{47}$$

Given Def. 40 of partial symmetric polynomials, we have

$$e_k^m = e_k^{m-1} + a_i\, e_{k-1}^{m-1} \tag{48}$$

Which is Eq. 44.

Recurrence condition 2: $e_k^{-i} = e_k^{|N|} - a_i\, e_{k-1}^{-i}$ **for** $(i \in N,\ 1 \le k \le |N|)$

Fix i in N, and fix a k s.t. $1 \le k \le |N|$. By Lemma X we have

$$e_k(N) = e_k(N - i) + a_i\, e_{k-1}(N - i) \tag{49}$$

Now, $e_k^{-i} := e_k(N-i)$ and $e_{k-1}^{-i} := e_{k-1}(N-i)$ (by def. leave-one-out symmetric polynomials). In addition, $e_k^{|N|} = e_k(N)$ (by definition of partial symmetric polynomials). Substituting these into the above equation we get:

$$e_k^{|N|} = e_k^{-i} + a_i\, e_{k-1}^{-i} \tag{50}$$

Subtract $a_i\, e_{k-1}^{-i}$ from both sides to get

$$e_k^{-i} = e_k^{|N|} - a_i\, e_{k-1}^{-i} \tag{51}$$

Which is Eq. 45.

This completes the proof. □

A.5 Proof - Complexity

Proposition 5 (Correctness and complexity of Algorithm 1) *Let (N, v) be a Noisy–OR game with $n = |N|$. Algorithm 1 computes the exact Shapley vector $\phi(v, N)$ as given in Proposition 2, and runs in $O(n^2)$ time and $O(n)$ space.*

Proof We match each stage of the algorithm to the analytic formula of Proposition 2 and then account for resources.

Complexity. Let $n = |N|$. Step 1 performs $\sum_{m=1}^{n} m = O(n^2)$ constant-time updates to the vector $e[0..n]$. In Step 3, for each player i the algorithm forms a temporary vector $e^-[0..n-1]$ using $n-1$ updates and then performs n further operations to accumulate ϕ_i; over all i this is again $O(n^2)$ work. No other loops depend quadratically on n. At any moment the algorithm stores only $e[\,]$, one temporary vector $e^-[\,]$, and the output $\phi[\,]$, all of length $O(n)$. Hence $T(n) = O(n^2)$ and $S(n) = O(n)$.

Correctness. From Proposition 2,

$$\phi_i = \frac{w_i}{n} \sum_{k=0}^{n-1} \frac{e_k^{(-i)}}{\binom{n-1}{k}},$$

where $e_k^{(-i)}$ are the leave–one–out elementary symmetric polynomials of Lemma 4. It therefore suffices to show that the algorithm computes (i) the full symmetric polynomials e_k, and (ii) for each i, the corresponding $e_k^{(-i)}$ used in the aggregation.

Step 1. The update

$$e[k] \leftarrow e[k] + a[m] \cdot e[k-1]$$

implements the recurrence $e_k^{(m)} = e_k^{(m-1)} + a_m e_{k-1}^{(m-1)}$ of Lemma 4. By induction on m, after the loop terminates we have $e[k] = e_k^{(n)} = e_k$ for all k.

Step 3 (per–i computation). For a fixed i, the algorithm initialises $e^-[0] = 1$ and applies

$$e^-[k] \leftarrow e[k] - a[i] \cdot e^-[k-1],$$

Algorithm 1 Noisy–OR Shapley value via elementary symmetric polynomials, $O(n^2)$ time and $O(n)$ space

---

Require: Weights $w[1..n] \in [0,1]$
Ensure: Shapley values $\phi[1..n]$

$a[j] \leftarrow 1 - w[j]$ for $j = 1..n$ (complements)

Step 1: Compute full elementary symmetric polynomials e_k

$e[0..n] \leftarrow 0;\ \ e[0] \leftarrow 1$ $(e_0 = 1)$
for $m = 1$ **to** n **do**
  for $k = m$ **down to** 1 **do**
    $e[k] \leftarrow e[k] + a[m] \cdot e[k-1]$ $(e_k^m = e_k^{m-1} + a_m e_{k-1}^{m-1})$
  end for
end for
Invariant: $e[k] = e_k$ for all $k = 0, \dots, n$.

Step 2: Precompute inverse binomial coefficients

for $k = 0$ **to** $n-1$ **do**
  $c[k] \leftarrow \dfrac{1}{\binom{n-1}{k}}$
end for

Step 3: For each i, compute e^{-i} on-the-fly and immediately aggregate ϕ_i

for $i = 1$ **to** n **do**
  $e^{-}[0] \leftarrow 1$ (temporary vector; reused each i)
  for $k = 1$ **to** $n-1$ **do**
    $e^{-}[k] \leftarrow e[k] - a[i] \cdot e^{-}[k-1]$ $(e_k^{-} = e_k - a_i e_{k-1}^{-})$
  end for
  $s \leftarrow 0$
  for $k = 0$ **to** $n-1$ **do**
    $s \leftarrow s + e^{-}[k] \cdot c[k]$
  end for
  $\phi[i] \leftarrow \dfrac{w[i]}{n} \cdot s$
end for
return ϕ

---

which is exactly the recurrence $e_k^{(-i)} = e_k - a_i e_{k-1}^{(-i)}$ with the correct boundary condition. Induction on k gives $e^{-}[k] = e_k^{(-i)}$ for all k. The algorithm then evaluates $\phi[i]$ which coincides with the closed-form expression above. Since this is done independently for each i, the returned vector equals $\phi(v, N)$.

This establishes correctness and the stated resource bounds. □